TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-powered Personalized Nutrition and Recommendation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Bio * Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software Solutions	Functional Restoration / Assistive & Welfare Devices	Daily Living Support Devices and Systems
Concept of Proposed Technology	This project represents a transformative step towards personalized nutrition and healthier eating. We propose a novel framework leveraging Generative Large Language Models (LLMs) and Knowledge Graphs (KGs) to develop an effective framework and chatbot App for personalized nutrition and food Recommendations to empower consumers with tailored advice and healthy food consumption.			
Rationale for Joint R&D Project	As the saying goes, "you are what you eat". Food is fundamental for human beings and eating well is essential to good health, yet it is hard to balance the dietary requirements between health and delicacy. Food also plays an essential role in cultural inheritance -- how we cook and eat food is an expression of cultural identity. Our focus is on helping health consumers find information related to specific food related healthcare goals, in a way that: i) supports their information seeking intention, ii) is tailored to their unique context, and iii) increases healthy behaviors via personalized			

	recommendation. The collaboration with Korean SMEs on this joint project will be crucial to realize our aim to recommend personalized and culturally aware diet advice and guidelines for healthy eating behaviors. For example, what would be the equivalent of the healthy "Mediterranean Diet" but using locally available ingredients and respecting the cuisine style, say in Korean cuisine?
Key Words	1. Healthy and Personalized Food Recommendation 2. Knowledge-graph enhanced Generative AI 3. Multimodal Food Representation Learning 4. Recipe Generation with Nutritional Constraints 5. Multilingual Food KG and Q&A Benchmark

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	Our core innovation is a knowledge-enhanced multi-modal and multi-lingual generative AI system that integrates LLMs with a comprehensive Food KG to provide tailored dietary guidance to consumers via a personalized App. We automate KG construction and continuously update it with diverse public data sources. Critically, the KG provides contextual grounding for the LLM, mitigating inaccuracies and enabling more reliable recommendations. The project comprises four key aims: (1) Food Representation Learning through multi-modal embeddings; (2) a Personalized Food Recommendation system incorporating user preferences and health constraints; (3) a Healthy Food Substitution module; and (4) Recipe Generation with Nutritional Constraints.			
R&D Target Specifications	We aim to develop novel approaches and assessments for the following components: 1. Personalized Conversation-based Recommendation 2. Knowledge Enhanced Recommendation and Generation 3. Enhanced Representation Learning and Healthy Food Substitution 4. Enhanced Food Knowledge Graph (Multi-modal and Multilingual, esp. focusing on English and Korean) The performance metrics will include state-of-the-art benchmark performance on personalized food recommendation tasks, substitution tasks, and nutritional estimation. Also, we will evaluate the system's alignment with human expectations via crowdsourcing and user studies.			

R&D Plan

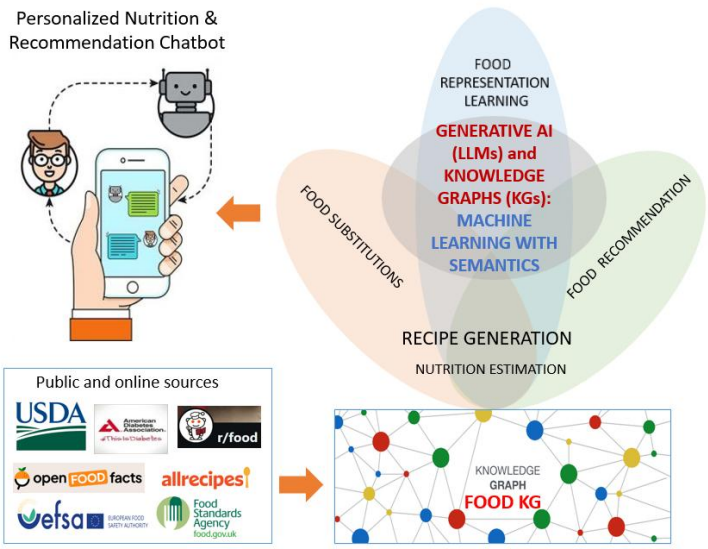

Figure 1. Personalized Nutrition and Recommendation App and Framework: Various public data sources, including USDA's FoodData Central, recipe websites, and online forums, feed into the creation of the Food Knowledge Graph (KG), which serves as the foundation for the tasks of food representation learning, personalized food recommendation, healthy food substitution, and recipe generation with nutritional constraints. An AI chatbot provides a conversational interface for personalized queries and responses, enabling users to interact with the system in natural language.

We propose four specific aims to develop a unified framework for personalized nutrition and healthy eating recommendations, as illustrated in Figure 1.

- Aim 1 – Food Representation Learning: Develop a multi-modal embedding model that captures semantic relationships between recipes, ingredients, nutrients, food images, and user-defined tags. This will involve deep learning methods to learn features from textual data and integrating them with image-based features to create effective latent space representations. This aim addresses the challenge of heterogeneous food data by creating a unified representation that can be used across all tasks in the project.
- Aim 2 – Personalized Food Recommendation: Design and implement a recommendation system that leverages the food knowledge graph and user preference data to provide a ranked list of food items tailored to individual needs. This will incorporate health constraints and dietary restrictions.
- Aim 3 – Healthy Food Substitution: Develop an ingredient substitution module that recommends alternatives to unhealthy ingredients while preserving the user preferences and nutritional value of recipes.
- Aim 4 – Recipe Generation with Nutritional Constraints: Create a recipe generation model that generates complete cooking steps and nutritional information based on user-defined queries and health goals, integrating user feedback to refine recommendations. This aim integrates the food knowledge graph with generative models to create functional and personalized recipes.

We anticipate a phased research and development approach, building foundational components in Year 1 (e.g., Aims 1 and 4, but also making progress on other aims), refining and integrating them in Year 2 (especially Aims 2 and 3), and focusing on prototype App development in Year 3 (realizing the vision in Figure 1).

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	We will bring our extensive world-class expertise on food computing [1] to collaborate effectively with the Korean SMEs to advance the main aims of this project on personalized food recommendation and nutrition. As part of the Health Empowerment through Analytics, Learning and Semantics project [2] (IBM-RPI collaboration), the PI led the creation of the largest open knowledge graph on recipes and their ingredients and nutrients – the FoodKG [3] – spanning over 67 million triples. We were among the first to propose the concept of a personalized health KG [4,5] focusing on the user-side as opposed to the clinicians. Our contribution includes the RECIPTOR [6] transformer architecture for food representation learning, with effective downstream performance on cuisine classification and other food computing tasks. We pioneered the first large-scale food recommendation system over the FoodKG in our pFoodReq framework [7] that employs a question-answering approach to help users elicit their food/dietary needs in terms of matching recipes. We further introduced the task of health-guided recipe recommendation over KGs [8,9], and an adversarial substitution approach to find semantically and contextually similar ingredients [10]. Our recent work has proposed state-of-the-art deep learning models such as LLaVA-Chef [11] for multi-modal generative model for recipes, and KERL [12] that uses KGs within a Retrieval Augmented Generation (RAG) framework to enhance LLM performance for personalized recipe recommendation, along with nutrition information and instructions. The PI also serves as an External Advisory Board member for the EU CoDiet Project [13] – an international research project that aims to combat diet-related diseases by addressing current gaps in our knowledge and developing new ways to personalize dietary advice and helping society to become healthier.
Expected R&R of Korean SME Partner	We would leverage the expertise of Korean SMEs for the following: 1. Creating a multilingual Food Knowledge Graph incorporating Korean foods 2. Developing a benchmark Q&A dataset creation for culturally aware food related questions 3. User testing to assess the efficacy of the personalized and health aware recommendations 4. Prototype App development for nutrition and recommendation chatbot 5. Integration with smart devices such as smart refrigerators (e.g., to scan fridge content to suggest healthy meals) 6. Overall guidance on translational aspects of our research prototypes for further commercialization

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Despite advances in food computing, achieving widespread healthy eating remains a significant challenge. Generic dietary guidelines often fail to account for individual variations in health status, dietary needs, cultural preferences, and access to affordable, nutritious foods. This mismatch contributes to rising rates of obesity, diet-related diseases, and inefficiencies within the agricultural supply chain. By enabling consumers with personalized food solutions, this project has the potential to enhance food security, improve nutrition, and promote sustainable agricultural practices by enabling tailored dietary advice and optimizing food consumption patterns. The resulting App will empower consumers, support local food systems, and potentially reduce healthcare costs.

6. References

(1) Gjorgjina Cenikj, Mauro Dragoni, Tome Eftimov, Barbara Korousic Seljak, Agnieszka Lawrynowicz, Fnu Mohbat, Oshani Seneviratne, Yoko Yamakata, and Mohammed J. Zaki. Neurosymbolic methods for food computing. In Handbook on Neurosymbolic AI and Knowledge Graphs, pages 1019–1056. IOS Press, 2025.

(2) https://idea.rpi.edu/research/projects/heals

(3) Steven Haussmann, Oshani Seneviratne, Yu Chen, Yarden Ne'eman, James Codella, Ching-Hua Chen, Deborah L McGuinness, and Mohammed J Zaki. FoodKG: semantics-driven knowledge graph for food recommendation. In The Semantic Web– ISWC: 18th International Semantic Web Conference, Auckland, New Zealand, pages 146–162. Springer, 2019.

(4) Nidhi Rastogi and Mohammed J. Zaki. Personal health knowledge graphs for patients. In Workshop on the Personal Health Knowledge Graph (with the Knowledge Graph Conference), May 2020.

(5) Ching-Hua Chen, Daniel Gruen, Jonathan Harris, James Hendler, Deborah L. McGuinness, Marco Monti, Nidhi Rastogi, Oshani Seneviratne, and Mohammed J. Zaki. Semantic technologies for clinically relevant personal health applications. In Personal Health Informatics: Patient Participation in Precision Health, Editors: Hsueh, Pei-Yun Sabrina and Wetter, Thomas and Zhu , Xinxin, pages 199–220, Cham, 2022. Springer International Publishing.

(6) Diya Li and Mohammed J. Zaki. RECIPTOR: An effective pretrained model for recipe representation learning. In ACM SIGKDD International Conference on Data Mining and Knowledge Discovery, Aug 2020.

(7) Yu Chen, Ananya Subburathinam, Ching-Hua Chen, and Mohammed J. Zaki. Personalized food recommendation as constrained question answering over a large-scale food knowledge graph. In Fourteenth ACM International Conference on Web Search and Data Mining (WSDM), Mar 2021.

(8) Diya Li, Mohammed J. Zaki, and Ching-Hua Chen. Nutrition guided recipe search via pre-trained recipe embeddings. In IEEE Workshop on Data Engineering Meets intelligent food and Cooking Recipes (DECOR), with IEEE ICDE Conference, Apr 2021.

(9) Diya Li, Mohammed J. Zaki, and Ching-Hua Chen. Health-guided recipe recommendation over knowledge graphs. Journal of Web Semantics, Special Issue on Knowledge Graphs and Information Retrieval, 75:100743, Jan 2023.

(10) Diya Li and Mohammed J. Zaki. Food knowledge representation learning with adversarial substitution. In Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing, AACL/IJCNLP - Volume 1: Long Papers, pages 653–664. Association for Computational Linguistics, Nov 2022.

(11) Fnu Mohbat and Mohammed J. Zaki. Llava-Chef: A multi-modal generative model for food recipes. In ACM Conference on Information and Knowledge Management (CIKM), Oct 2024.

(12) Fnu Mohbat and Mohammed J. Zaki. KERL: Knowledge-enhanced personalized recipe recommendation using large language models. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL Volume 1: Long Papers), Jul 2025.

(13) https://www.codiet.eu/advisory-board/mohammed-j-zaki/

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization